Thermodynamic description of the Ag–Ni system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Ag–Ni binary system as calculated with pycalphad from the SGTE dataset AgNi-04Din.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L2 → L1 + (Ni), lies at 1703.4 K.

1. Introduction

The Ag–Ni system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as miscibility gap, monotectic. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of A.T. Dinsdale, unpublished, 2004. [2]. The calculations use pycalphad 0.11.2 at 1 atm.

Pure Ag melts at 1234.9 K. Pure Ni melts at 1728.2 K.

2. Phase diagram and invariant reactions

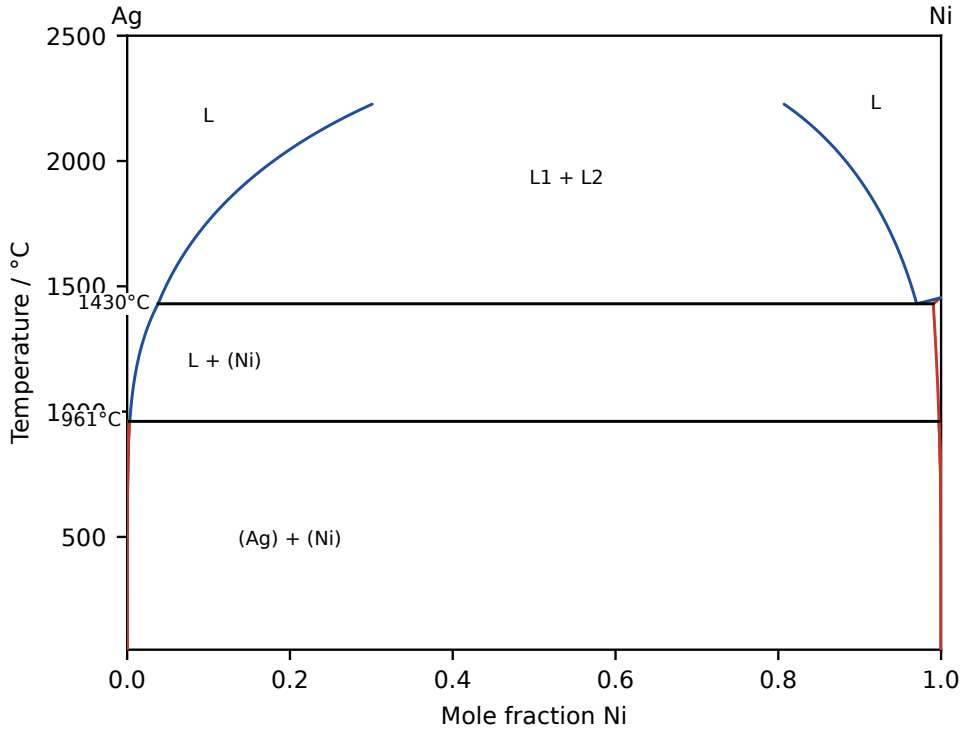


Fig. 1. Calculated Ag–Ni phase diagram.

At 1703.4 K the monotectic L2 → L1 + (Ni) takes place; the L1 phase then contains 3.8 at.%. We obtain 1234.2 K for the eutectic reaction (L → (Ag) + (Ni)), where L has 0.3 at.% Ni.

Table 1. Invariant reactions and congruent points, calculated (Ag–Ni).

Reaction	Type	T / K	Compositions (x(Ni))		
		this work	phase 1	phase 2	phase 3
L2 → L1 + (Ni)	monotectic	1703.4	0.970	0.038	0.990
L → (Ag) + (Ni)	eutectic	1234.2	0.003	0.003	0.997

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Ag–Ni).

	Structure			Stability range / K		Homogeneity range	
Phase	Pearson	Strukturb.	Prototype	from	to	x(Ni)	at T / K
(Ag)	cF4	A1	Cu	< 300	1234.9	0.000–0.002	1230
(Ni)	cF4	A1	Cu	< 300	1728.2	0.990–1.000	1700

According to Table 2, (Ag) remains stable up to 1234.9 K (melting of the pure element). The solubility of Ag in (Ni) reaches 1.0 at.% at 1703.4 K.

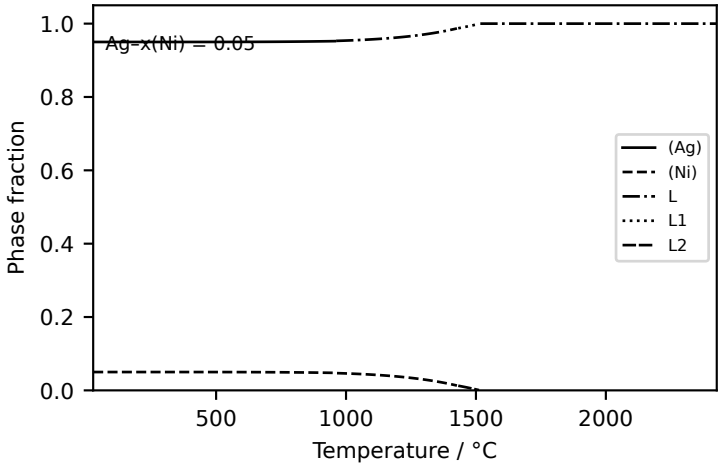


Fig. 2. Calculated equilibrium phase fractions versus temperature for Ag–x(Ni) = 0.05.

4. Thermodynamic properties

At 2600 K the integral mixing enthalpy of the liquid has its maximum, 15118 J/mol, at 38.0 at.% Ni. For the equiatomic liquid at 2600 K, the Gibbs energy of mixing is −3.88 kJ/mol-atom.

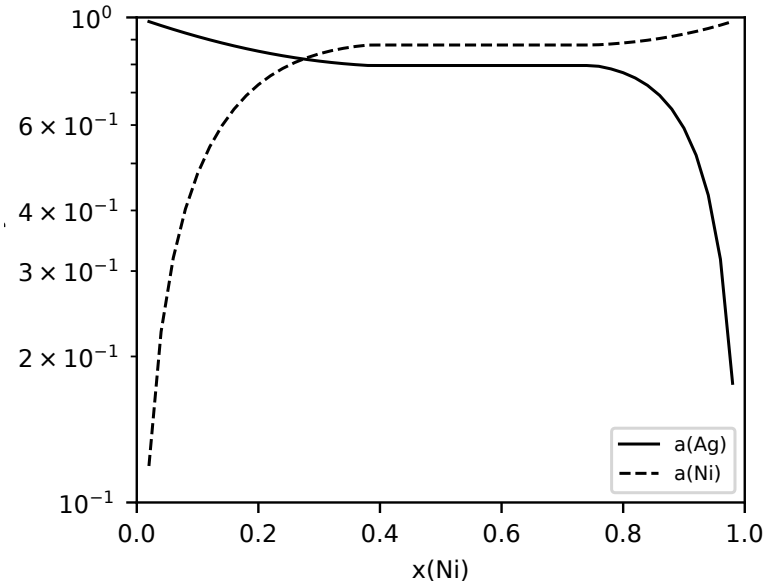


Fig. 3. Calculated activities in the liquid at 2600 K (reference: pure liquids).

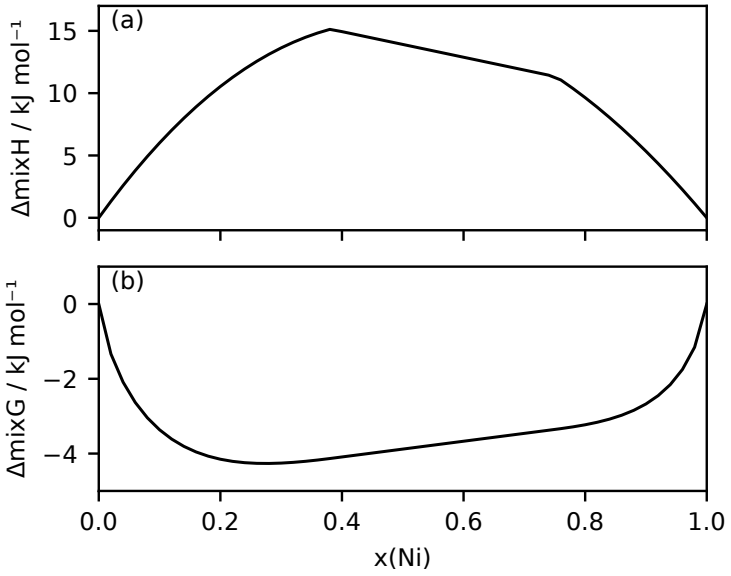


Fig. 4. Calculated (a) mixing enthalpy and (b) mixing Gibbs energy of the liquid at 2600 K.

Table 3. Redlich–Kister parameters of the liquid (Ag–Ni).

	Liquid: L(ν) = a + b·T + c·T·ln T		
ν	**a**	**b**	**c**
L0	63052.65	−7.119265	0.0
L1	4654.55	−3.636835	0.0

Table 4. Integral and partial mixing properties of the liquid at 2600 K (Ag–Ni).

x(Ni)	ΔmixH / kJ mol⁻¹	ΔmixG / kJ mol⁻¹	a(Ag)	a(Ni)
0.100	6.01	−3.36	0.9134	0.4764
0.300	13.63	−4.25	0.8128	0.8415
0.500	13.91	−3.88	0.7960	0.8773
0.700	11.85	−3.46	0.7960	0.8773
0.900	5.34	−2.67	0.5912	0.9241

Reference states: pure liquid Ag and Ni at 2600 K; per mole of atoms.

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] A.T. Dinsdale, unpublished, 2004.

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.